Downloaded from geology.gsapubs.org on October 28, 2010

Geology

Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from the Carolina slate belt, Albemarle, North Carolina: Implications for terrane amalgamation with North America

J. R. Noel, D. J. Spariosu and R. D. Dallmeyer

Geology 1988;16;64-68
doi: 10.1130/0091-7613(1988)016<0064:PAAAAF>2.3.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geology

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from geology.gsapubs.org on October 28, 2010

Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from the Carolina slate belt, Albemarle, North Carolina: Implications for terrane amalgamation with North America

J. R. Noel, D. J. Spariosu, R. D. Dallmeyer Department of Geology, University of Georgia, Athens, Georgia 30602

ABSTRACT

Paleomagnetic and $^{40}Ar/^{39}Ar$ dating studies of Late Proterozoic metamorphic rocks from the Carolina slate belt were undertaken to gain insight into the history of this suspect lithostratigraphic terrane. Geochronological data suggest a peak metamorphic age of ca. 455 Ma and a later thermal overprint at ca. 185 Ma. Thermal demagnetization indicates two magnetization components. A low-temperature component yields a pole close to 180–190 Ma segments of the North American apparent polar wander path. A high-temperature component postdates folding and is inferred to have been acquired at ca. 455 Ma (corresponding paleomagnetic pole is lat 20°N, long 80°E). Although paleomagnetic data from the Carolina slate belt remain too ambiguous for construction of a unique polar wander path, consideration of available paleolatitudes suggests that the terrane may have been proximal to cratonic North America after ca. 455 Ma. This result contrasts markedly with high southerly Ordovician paleolatitudes inferred from paleomagnetic studies within central and northern Appalachian terranes.

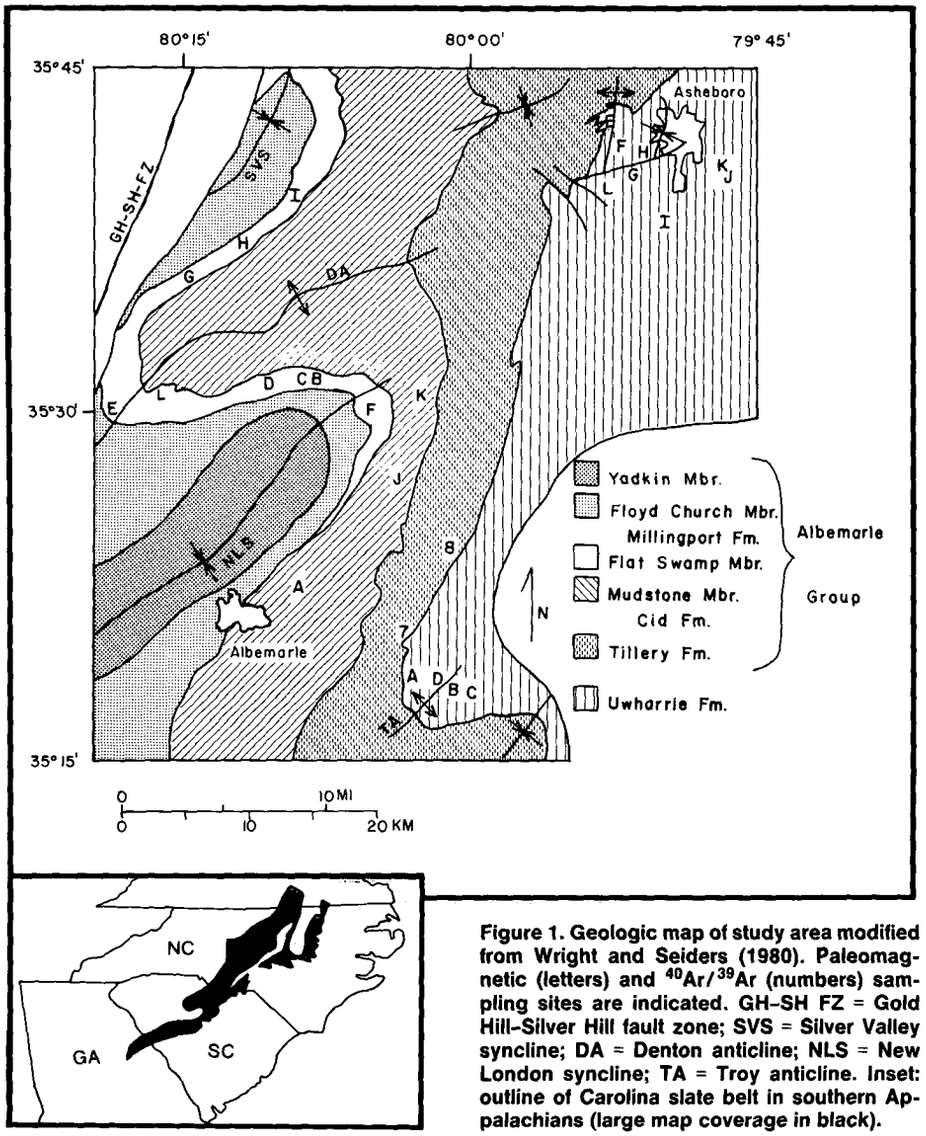


Figure 1. Geologic map of study area modified from Wright and Seiders (1980). Paleomagnetic (letters) and $^{40}Ar/^{39}Ar$ (numbers) sampling sites are indicated. GH–SH FZ = Gold Hill–Silver Hill fault zone; SVS = Silver Valley syncline; DA = Denton anticline; NLS = New London syncline; TA = Troy anticline. Inset: outline of Carolina slate belt in southern Appalachians (large map coverage in black).

INTRODUCTION

The Carolina slate belt contains a distinctive suite of Late Proterozoic–Lower Cambrian, variably deformed and metamorphosed, volcanic and volcaniclastic sedimentary rocks that extends from central Georgia to southern Virginia (Fig. 1). Paleontologic characteristics indicate that at least parts of the Carolina slate belt originated in a setting removed from North America (Secor et al., 1983). Williams and Hatcher (1982) included the Carolina slate belt in an "Avalon terrane" which they extended the length of the Appalachian orogen. Even broader correlations have been suggested by Rast et al. (1976) and Rast and Skehan (1983) who included the Carolina slate belt within an extensive microcontinent (Avalonia) which also included sequences now found as exotic terranes in Great Britain, western Europe, and Africa. On the other hand, Zen (1983) and Keppie (1987) argued that the Avalon terrane is composite and is represented by several individually distinct terranes, each with unique stratigraphic and/or accretionary histories. Although paleomagnetic data from parts of the northern and central Appalachians (Van der Voo and Johnson, 1985) imply high southern latitudes during Ordovician time for several eastern Appalachian sequences (including the Avalon terrane in Nova Scotia), these data are inconclusive as to whether these sequences (and by implication the Carolina slate belt) were part of a larger continental plate that moved as a coherent tectonic unit or were independent terranes that were amalgamated before and/or during collision with Laurentia.

In an attempt to resolve this question, integrated paleomagnetic and $^{40}Ar/^{39}Ar$ dating studies have been carried out in the Carolina slate belt near Albemarle, North Carolina. Results are presented here, and they indicate that in this area the Carolina slate belt represents a structural unit with a Paleozoic history distinct from other tectonic units that have been included in the Avalon terrane.

GEOLOGIC SETTING

The study area is located in central North Carolina (Fig. 1). Descriptions of the geologic

Note: Additional material for this article is Supplementary Data 8802, available on request from the GSA Documents Secretary (see footnote 1).

setting are provided in Conley and Bain (1965), Conley (1962), Stromquist et al. (1971), Stromquist and Sundelius (1969), Sundelius (1970), Sundelius and Stromquist (1978), Seiders (1981), and Harris and Glover (1985). The oldest exposed rocks are volcaniclastic units of the Uwharrie Formation. These rocks are conformably overlain by metasiltstones (Tillery Formation) at the base of the Albemarle Group; the metasiltstones are overlain by the Cid Formation, which consists of a lower metamudstone unit (Mudstone Member) and an upper volcanic sequence (Flat Swamp Member). Upper parts of the Albemarle Group are represented by metasandstone and metagraywacke. Both the Uwharrie Formation and the Albemarle Group are folded along northeast-trending, subvertical axial surfaces. This was accompanied by development of a variably penetrative cleavage that formed concomitantly with a low to middle greenschist facies regional metamorphism. Kish et al. (1979) reported a whole-rock K-Ar age of 483 ± 15 Ma for slate within the Albermarle Group and interpreted this to date postmetamorphic cooling through temperatures required for intracrystalline retention of argon in constituent fine-grained white micas. Partly on the basis of this date, Glover et al. (1983) suggested a Late to early Middle Ordovician age for the folding and metamorphism observed in both the Uwharrie Formation and the Albemarle Group.

The area was later affected by faulting and intrusion of numerous Mesozoic mafic dikes. Although the unmetamorphosed diabase dikes in the Albemarle area have not been isotopically dated, they are similar to Triassic-Jurassic dikes in other parts of the crystalline Appalachians that have been related to initial rifting associated with opening of the present Atlantic Ocean (Dallmeyer, 1975; Deininger et al., 1975; Sutter and Smith, 1979; Dooley and Smith, 1982).

SAMPLING AND ANALYTICAL METHODS

A total of 201 oriented drill cores were collected from 13 sites in the Uwharrie Formation, 11 sites in the Cid Formation, and 1 site in an unmetamorphosed diabase dike intruding the Cid Formation (Fig. 1). Three of the sites in the Cid Formation were pyritiferous volcanogenic metasiltstones. All others were volcanic or volcaniclastic lithologies. A suite of slate samples was collected within the Tillery Formation for $^{40}Ar/^{39}Ar$ geochronology.

Thin-section examination of the paleomagnetic samples reveals roughly equal numbers of mafic and felsic samples among the metavolcanic rocks (details in Noel, 1986). Conventional paleomagnetic laboratory procedures were employed, including alternating field and thermal demagnetization procedures. Thermal demagnetization was used for most of the samples because it proved more effective in separating multiple magnetization components and also because the resulting data are more directly comparable to argon release data. Directions were determined by using principal component analysis (Kirschvink, 1980) over a minimum of three demagnetization steps determined from examination of vector end-point diagrams. Component directions with maximum angular deviations >15° were not considered further. Isothermal remanent magnetization acquisition studies were performed on five selected samples to aid in determination of magnetic mineralogy.

The techniques used during $^{40}Ar/^{39}Ar$ analysis followed those described in detail by Dallmeyer and Keppie (1987). Variations in the flux of neutrons along the length of the irradiation assembly were monitored with several mineral standards, including MMhb-1 (Alexander et al., 1978). Apparent $^{40}Ar/^{39}Ar$ ages were calculated from the corrected isotopic ratios by using the decay constants and isotopic abundance ratios listed by Steiger and Jäger (1977). Two-sigma, intralaboratory uncertainties are reported. Apparent K/Ca ratios may be calculated through the relation 0.518 (±0.005) × ($^{39}Ar/^{37}Ar$) corrected.

TABLE 1. SITE MEAN DIRECTIONS AND PALEOMAGNETIC POLES

Site	n/N	T_b(°C)	Geographic Dec	Geographic Inc	Corrected Dec	Corrected Inc	k	95	Pole Lat	Pole Long	dp, dm
L Component											
UWK	5/5	<440	9.0	31.8	359.0	8.0	63.0	9.7			
CIDL	4/4	<435	15.6	30.9			166.0	7.2			
MEAN	9/9		12.0	31.4			86.4	5.6	68.6	67.1	3.5, 6.3
H Component											
UWC	5/6	450-500	12.6	-39.7	0.7	-17.9	61.5	9.8			
UWC	6/6	500-580	189.8	47.4	175.3	24.2	186.9	4.4			
UWJ	4/6	>450	251.4	24.9	260.1	6.6	79.9	10.4			
CIDF	3/7	>450	220.4	48.6	233.4	39.6	29.4	23.2			
MEAN	18/25		220.9	40.0			10.3	14.2	20.1	60.0	10.3, 17.
	18/25				220.5	23.5	4.7	22.6	-	-	- -

Note: Poles calculated for site location at lat 35.5° N, long 280° E; n/N is number of samples used for site mean/ number of samples demagnetized.

RESULTS

Paleomagnetism

Natural remanent magnetization (NRM) intensities of the samples range from 10^{-5} A/m to 10^{-1} A/m; 80% have intensities $<10^{-3}$ A/m. NRM directions are highly scattered but show a predominance of positive inclinations. No significant differences were noted between the Uwharrie and Cid Formations.

Stable magnetization components proved difficult to isolate in many of the samples; however, two consistent and stable magnetization components are evident in samples from six sites (Table 1). The first is consistently observed in samples from both the unmetamorphosed diabase dike and in one site from the Uwharrie Formation (UWK). Except for varying amounts of viscous overprint of present field direction (removed by 100–200 °C), these samples possess only one well-defined L component with a north-northeasterly declination and intermediate downward inclinations (Fig. 2C). In all samples this L magnetization is removed below 440–470 °C. The only magnetization stable to high temperatures is an H magnetization observed in samples from sites UWC, UWJ (Uwharrie Formation), and CIDF (Cid Formation). This direction is characterized by southwest declinatioons and moderately positive inclinations in the 450–580 °C range. However, behavior below 450 °C differs considerably between sites. CIDF samples are weakly magnetized but show univectorial decay in only the H direction from NRM 540 °C (Fig. 2A). By contrast, UWJ samples contain a westerly, shallow direction until 450 °C (above which only the H direction is present). UWC samples behave more complexly (Fig. 2B). From NRM 200 °C a northerly direction with positive inclinations is removed. This either represents a viscous remanent magnetization (VRM) or an L direction overprint. Between 200 and 400 °C a composite direction or unresolvable component is observed. From 400 to 500 °C a northerly, up direction is removed. This is followed by reversal and decay to the origin from the H direction to 580 °C. We interpret this as a dual polarity expression of the H component magnetization.

A possible third magnetization is observed in six samples from site CIDK within the Cid Formation. This component decays univectorially to the origin and is completely unblocked at 400 °C. No stable magnetization can be discerned above 400 °C. Because of the low blocking temperature and the absence of any corroboration of the direction in other sites, this magnetization is not considered further.

In no samples is there evidence for stable magnetization components above 580 °C. Little magnetization intensity was usually left at high-temperature demagnetization steps. These characteristics, along with isothermal remanent magnetization (IRM) acquisition experiments on five selected samples, indicate that magnetite is the remanence carrier for each of the identified magnetization components.

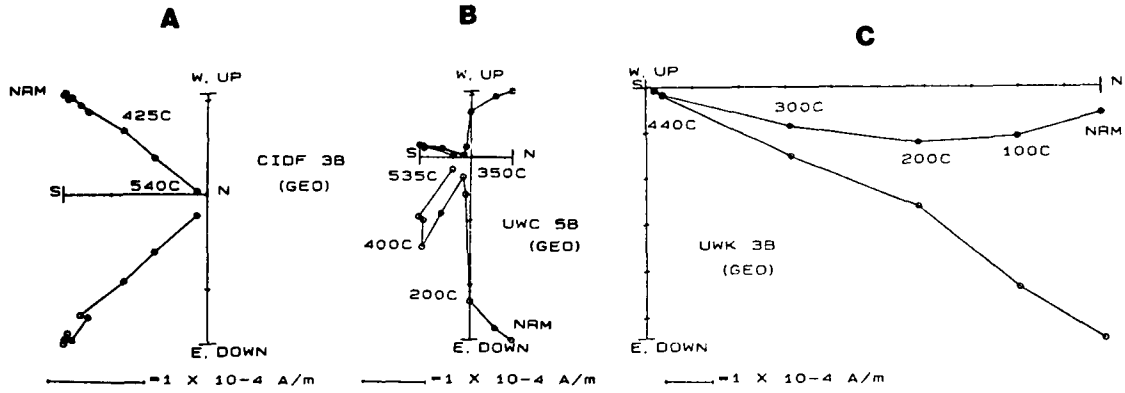


Figure 2. Orthogonal plots of thermal demagnetizations of representative samples from Uwharrie and Cid Formations. Solid symbols are projections on horizontal plane; open symbols, vertical plane. Demagnetization temperatures in °C. A and B illustrate different behaviors of H component. C is example of demagnetization behavior of L component.

Geochronology

Two slate samples were collected from the Tillery Formation for whole-rock, $^{40}Ar/^{39}Ar$ incremental-release dating. These samples consist predominantly of fine-grained white mica that is intergrown with subordinate chlorite and aligned along a penetrative cleavage. Trace amounts of detrital quartz and feldspar occur within the samples. The $^{40}Ar/^{39}Ar$ analytical data are listed in Table 2[1] and are displayed as age and apparent K/Ca spectra in Figure 3.

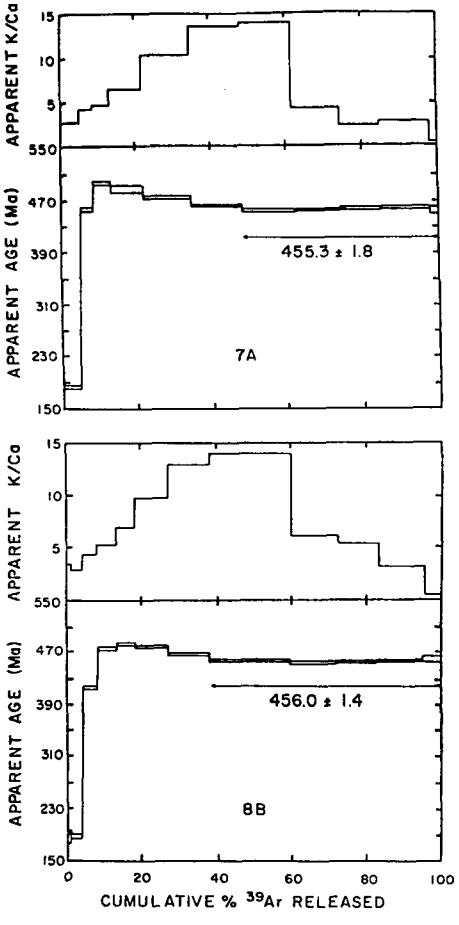


Figure 3. Incremental-release $^{40}Ar/^{39}Ar$ age and apparent K/Ca spectra for whole-rock slate samples from Tillery Formation, Albemarle area, North Carolina. Two-sigma intralaboratory uncertainties in apparent ages indicated by vertical width of bars. Experimental temperatures increase from left to right. Plateau ages listed on each spectrum.

The two whole-rock samples display internally discordant $^{40}Ar/^{39}Ar$ age spectra. These are of generally similar character and yield total-gas ages of 450.4 ±2.3 Ma (sample 7A) and 450.1 ±2.0 Ma (sample 8B). In each, low-temperature gas fractions (450–475 °C) yield anomalously young ages of ca. 180–185 Ma. These increase markedly to ca. 476–497 Ma in the 525 °C increments. Apparent ages systematically decrease in the 550–600 °C increments and remain mutually similar throughout intermediate- and high-temperature parts of both analyses. Plateau ages of 455.3 ±1.8 Ma (sample 7A) and 456 ±1.4 Ma (sample 8B) are recorded. Fusion increments of both samples yield anomalously old apparent ages. The $^{40}Ar/^{36}Ar$ vs. $^{39}Ar/^{36}Ar$ isochron treatment of the plateau data from each sample yields ages of 453.0 ±3.7 Ma (MSWD = 0.21; ordinate intercept = 382.3 ±14.1: sample 7A) and 457.8 ±1.2 Ma (MSWD = 0.61; ordinate intercept = 210 ±4.7: sample 8B).

Apparent K/Ca ratios are relatively low in the small gas increments evolved from both samples at low experimental temperatures. Markedly higher and relatively constant values are recorded in the larger gas increments evolved at intermediate temperatures. Systematically lower ratios are recorded in increments evolved during high-temperature portions of both analyses.

INTERPRETATION

Geochronology

The two whole-rock slate samples display similarly discordant $^{40}Ar/^{39}Ar$ age spectra. Although the samples consist primarily of fine-grained white mica, systematic intra-sample variations in apparent K/Ca ratios suggest that several other phases contributed gas at various stages in the whole-rock analyses. Relative to the white mica, these phases appear to have included (1) a nonretentive phase with a low K/Ca ratio and (2) a much less abundant and more refractory phase with a low K/Ca ratio. Mineralogical characteristics and observed modal variations suggest that these phases are chlorite and detrital feldspar, respectively. It is proposed that most of the observed spectra discordance is related to differences in the modal abundance of these phases relative to the white mica (see Reuter and Dallmeyer, 1987).

Both samples display internally discordant $^{40}Ar/^{39}Ar$ spectra; however, plateaus are defined by most of the intermediate- and high-temperature increments. These likely relate to gas evolved from white mica and are interpreted to closely date the last cooling through temperatures required for intracrystalline retention of argon (325 ±25 °C). The slightly older apparent ages recorded in the 525–600 °C increments are considered to reflect experimental evolution of extraneous argon components from relatively nonretentive sites in chlorite. The anomalously older dates recorded in the fusion increments likely reflect evolution of gas from feldspar. The 180–185 Ma apparent ages recorded in the 450–475 °C increments are intriguing because this pattern of spectra discordance is similar to that reported to characterize whole-rock slate systems affected by a mild geologic reheating following initial postmetamorphic cooling through argon retention temperatures (e.g., Dallmeyer et al., 1986; Keppie and Dallmeyer, 1987). The 180–185 Ma ages are similar to those reported for other diabase dikes in the southern and central Appalachians and may reflect thermal effects related to their emplacement. Constituent white micas could have suffered a minor diffusive loss of argon during the reheating. The expelled Ar could have migrated into chlorite and be manifested in the $^{40}Ar/^{39}Ar$ whole-rock analyses by the slightly older apparent ages observed in the 525–600 °C increments (see Dallmeyer and Keppie, 1987).

Paleomagnetism

Stable magnetization directions are plotted in Figure 4. Although no fold test is possible on the

[1]Table 2, Geological Society of America Supplementary Material 8802, is available on request from Documents Secretary, Geological Society of America, P.O. Box 9140, Boulder, CO 80301.

L magnetization, several features suggest it is of Triassic-Jurassic age: (1) agreement between the L direction in the metavolcanics and in the Mesozoic dikes; (2) agreement between these directions and those reported by Dooley and Smith (1982) for dikes in South Carolina; and (3) the calculated paleopole (68.6°N lat, 67.1°E long; semiaxes of oval of confidence [dp, dm] = 3.5, 6.3) is within the North American apparent polar wander path (APWP) envelope for the 170–200 Ma interval. The L magnetization is therefore interpreted to have formed during the 185 Ma thermal overprint suggested by the $^{40}Ar/^{39}Ar$ results.

The H magnetization direction is defined in 12 samples from 3 sites and fails the fold test at the 95% confidence level (k1/k2 = 2.19; i.e., the ratio of precision parameter before [k1] and after [k2] unfolding). Because the unblocking temperatures correspond fairly well to the temperatures of peak metamorphism and the $^{40}Ar/^{39}Ar$ results imply that the rocks cooled through ~325 °C at ca. 455 Ma, the age of the magnetization is inferred to postdate folding and predate ca. 455 Ma. Because folding and metamorphism appear to have occurred concomitantly, the cooling probably corresponds to a relatively short Ordovician interval. Studies of relaxation time vs. time held at elevated temperatures (e.g., Pullaiah et al., 1975) imply that magnetization acquired during slow cooling over tens of millions of years from peak metamorphic (350–550 °C) down to 325 °C could correspond to 450–580 °C laboratory blocking temperatures. In the absence of evidence of subsequent remagnetization events prior to the Mesozoic partial thermal remanent magnetization (pTRM), we infer an Ordovician age for the H magnetization.

DISCUSSION

Representative paleopoles for Ordovician North America and the Carolina slate belt are shown in Figure 5. Ellwood and Secor (1985) obtained preliminary paleomagnetic directions from metaquartzites collected in the Carolina slate belt near Columbia, South Carolina. Although scattered and representing only 4 sites, the metaquartzites yielded a postfolding magnetization that, based on $^{40}Ar/^{39}Ar$ geochronology in the area (Dallmeyer et al., 1986), was inferred to represent a Devonian magnetization age. However, the calculated paleopole (43°N, 63°E) does not agree with middle–late Paleozoic segments of the North American APWP. Vick et al. (1985) reported a paleopole at 35°N, 118°E obtained from metasedimentary rocks of the Carolina slate belt collected within Union County, North Carolina. A Devonian age was inferred for this pole on the basis of its proximity to Late Devonian–Early Carboniferous poles from North America and the Avalon terrane of the northern Appalachians. The H direction of this study appears to yield an Ordovician paleopole at 20°N, 60°E. A paleomagnetic study of Early Carboniferous plutons from the Carolina slate belt (Barton and Brown, 1983) gave a pole near the late Paleozoic portion of the North American APWP. These plutons are located in areas affected by Alleghanian tectonothermal events (Glover et al., 1983).

There is a marked disparity between the various paleomagnetic studies carried out in the Carolina slate belt. The pole reported by Vick et al. (1985) falls within the Late Ordovician envelope of the North American APWP. Considering the proximity of the areas and the similarity of the rocks used in their study to the Tillery Formation slate sampled for our $^{40}Ar/^{39}Ar$ experiments, it is quite possible that their result also represents an Ordovician magnetization. This would suggest no large separation between cratonic North America and this portion of the Carolina slate belt following the Middle Ordovician. The region around Columbia, South Carolina, studied by Ellwood and Secor (1985) underwent deformation and metamorphism during both the Devonian and the Permian (Dallmeyer et al., 1986). The pole obtained by Ellwood and Secor is, nonetheless, anomalous with respect to any part of the North American APWP.

Studies by Ellwood and Secor (1985) and Vick et al. (1985), together with this study, yield paleolatitudes of 7°S, 22°S, and 23°S. Within error limits, these paleolatitudes are not significantly different from those inferred for adjacent locations on the North American craton for the Ordovician through the Devonian (Scotese, 1984). They are considerably lower than paleolatitudes reported for the Upper Ordovician Dunn Point Formation of Avalonian Nova Scotia (42°S; Van der Voo and Johnson, 1985), and the 62°S and 81°S paleolatitudes of the Arden pluton (Rao and Van der Voo, 1980) and the Wissahocken Gneiss (Brown and Van der Voo, 1983) of the Maryland-Delaware Piedmont. These, along with similar high paleolatitudes from parts of Europe and North Africa, have been cited as evidence for an Armorican continent closely associated with Gondwana during

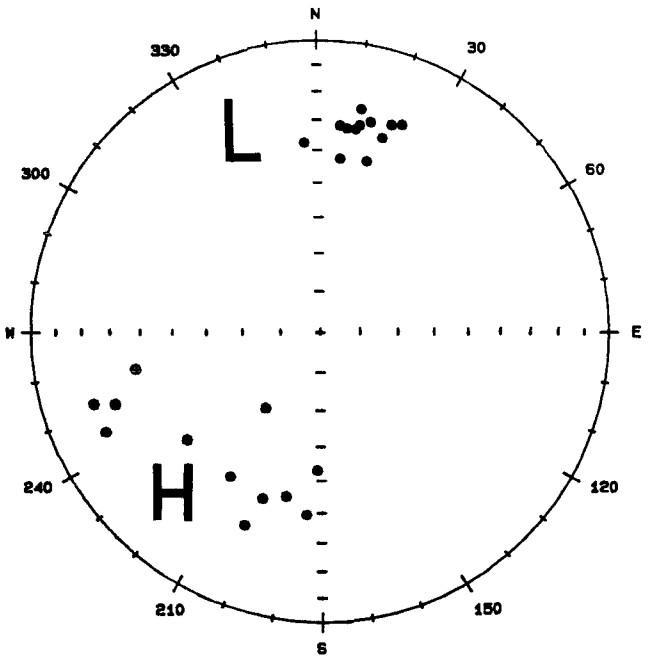


Figure 4. Equal-area projection of stable end-point magnetization directions from Cid and Uwharrie Formations. All points are plotted on lower hemisphere; normal polarity directions from site UWC (Table 2; see footnote 1) are inverted for clarity.

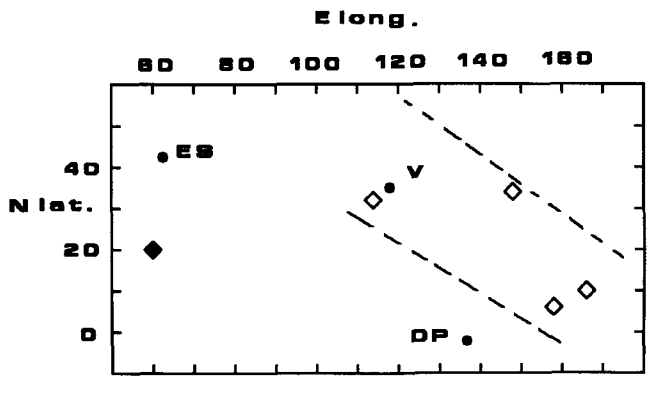


Figure 5. Paleomagnetic (north) poles used for comparison. Solid diamond represents pole from H magnetization in this study. Open diamonds represent reliable paleopoles from Ordovician of cratonic North America, and dashed-line envelope represents probable trend of North American apparent polar wander path from lower right to upper left (poles from Van der Voo and French, 1977; Watts and Van der Voo, 1979; Van der Voo et al., 1976; Jackson and Van der Voo, 1985). Solid circles are paleopoles from other localities in Carolina slate belt (ES = Ellwood and Secor, 1985; V = Vick et al., 1985) and from Avalon terrane of Nova Scotia (DP = Van der Voo and Johnson, 1985).

the Ordovician (Perroud et al., 1984). It is unlikely that the parts of the Carolina slate belt investigated in the present study had similar associations.

The $^{40}Ar/^{39}Ar$ whole-rock slate results from the Albemarle area of the Carolina slate belt are in marked contrast to those reported by Dallmeyer et al. (1986) from similar sequences near Columbia, South Carolina. In that area, whole-rock slate from northwesternmost parts of the Carolina slate belt record a ca. 340–360 Ma metamorphism. Southeastward across the Carolina slate belt the whole-rock systems are systematically overprinted by a ca. 280–290 Ma tectonothermal event which is more penetratively recorded in the Kiokee belt. The Albemarle area does not appear to have been affected by either the Devonian or the Permian events observed in the Columbia transect. Whether this represents different accretion times for separate terranes or reflects a pattern of promontories and reentrants in Acadian and Alleghanian metamorphic fronts remains unclear. Nonetheless, by strict definition, South Carolina and North Carolina parts of the Carolina slate belt should be viewed as distinct and separate terranes, qualifying on the basis of contrasting tectonothermal histories.

REFERENCES CITED

Alexander, E.C., Jr., Michelson, G.M., and Lanphere, M.A., 1978, MMhb-1: A new $^{40}Ar/^{39}Ar$ dating standard, *in* Zartman, R.E., ed., Fourth international conference on geochronology, cosmochronology, and isotope geology: U.S. Geological Survey Open-File Report 78-701, p. 6–8.

Barton, C., and Brown, L., 1983, Paleomagnetism of Carboniferous intrusions in North Carolina: Journal of Geophysical Research, v. 88, p. 2327–2335.

Brown, P.M., and Van der Voo, R., 1983, A paleomagnetic study of piedmont metamorphic rocks from northern Delaware: Geological Society of America Bulletin, v. 94, p. 815–822.

Conley, J.F., 1962, Geology of the Albemarle quadrangle, North Carolina: North Carolina Department of Conservation and Development and the Division of Mineral Resources Bulletin 75, 26 p.

Conley, J.F., and Bain, G.L., 1965, Geology of the Carolina Slate belt west of the Deep River–Wadesboro Triassic Basin, North Carolina: Southeastern Geology, v. 6, p. 117–138.

Dallmeyer, R.D., 1975, The Palisades sill: A Jurassic intrusion? Evidence from $^{40}Ar/^{39}Ar$ incremental release ages: Geology, v. 3, p. 243–245.

Dallmeyer, R.D., and Keppie, J.D., 1987, Polyphase late Paleozoic tectonothermal evolution of the southwestern Meguma terrane, Nova Scotia: Canadian Journal of Earth Sciences.

Dallmeyer, R.D., Wright, J.E., Secor, D.T., Jr., and Snoke, A.W., 1986, Character of the Alleghanian orogeny in the southern Appalachians: Part II. Geochronological constraints on the tectonothermal evolution of the eastern Piedmont in South Carolina: Geological Society of America Bulletin, v. 97, p. 1329–1334.

Deininger, R.W., Dallmeyer, R.D., and Neathery, T.L., 1975, Chemical variations and K-Ar ages of diabase dikes in east-central Alabama: Geological Society of America Abstracts with Programs, v. 7, p. 482.

Dooley, R.E., and Smith, W.A., 1982, Age and magnetism of diabase dykes and tilting of the Piedmont: Tectonophysics, v. 90, p. 283–307.

Ellwood, B.B., and Secor, D.T., Jr., 1985, Evidence for remagnetization of the Carolina Slate Belt trilobite locality: EOS (American Geophysical Union Transactions), v. 66, p. 257.

Glover, L., III, Speer, J.A., Russell, G.S., and Farrar, S.S., 1983, Ages of regional metamorphism and ductile deformation in the central and southern Appalachians: Lithos, v. 16, p. 223–245.

Harris, C.W., and Glover, L., III, 1985, The Virgilina deformation: Implications of stratigraphic correlation in the Carolina Slate Belt: Carolina Geological Society Field Trip Guide, 58 p.

Jackson, M., and Van der Voo, R., 1985, A Lower Ordovician paleomagnetic pole from the Oneota Dolomite, upper Mississippi River valley: Journal of Geophysical Research, v. 90, p. 449–461.

Keppie, J.D., 1987, The Appalachian collage: Uppsala, International Geological Correlation Program Caledonide Orogen Volume.

Keppie, J.D., and Dallmeyer, R.D., 1987, Dating transcurrent terrane accretion: An example from the Meguma and Avalon composite terranes in the northern Appalachians: Tectonics.

Kirschvink, J.K., 1980, The least-squares line and plane and the analysis of paleomagnetic data: Royal Astronomical Society Geophysical Journal, v. 62, p. 699–718.

Kish, S.T., Butler, J.R., and Fullagar, P.D., 1979, The timing of metamorphism and deformation in the central and eastern Piedmont of North Carolina: Geological Society of America Abstracts with Programs, v. 11, p. 184–185.

Noel, J.R., 1986, Paleomagnetism of Precambrian metavolcanic rocks from the Carolina Slate Belt [M.S. thesis]: Athens, University of Georgia, 129 p.

Perroud, H., Van der Voo, R., and Bonhommet, N., 1984, Paleozoic evolution of the Armorica plate on the basis of paleomagnetic data: Geology, v. 12, p. 579–582.

Pullaiah, G., Irving, I., Buchan, K.L., and Dunlop, D.J., 1975, Magnetization changes caused by burial and uplift: Earth and Planetary Science Letters, v. 28, p. 133–143.

Rao, K.V., and Van der Voo, R., 1980, Paleomagnetism of a Paleozoic anorthosite from the Appalachian piedmont, northern Delaware: Possible tectonic implications: Earth and Planetary Science Letters, v. 47, p. 113–120.

Rast, N., and Skehan, J.W., 1983, The evolution of the Avalonian plate, *in* Friedman, M., and Toksoz, M.N., eds., Continental tectonics: Structure, kinematics and dynamics: Tectonophysics, v. 100, p. 257–286.

Rast, N., O'Brien, R.H., and Wardle, R.J., 1976, Relationships between Precambrian and lower Paleozoic rocks of the 'Avalon Platform' in New Brunswick, the northeast Appalachians and the British Isles: Tectonophysics, v. 30, p. 315–338.

Reuter, A., and Dallmeyer, R.D., 1987, Significance of $^{40}Ar/^{39}Ar$ age spectra of whole-rock and constituent grain size fractions from anchizonal slates: Tectonics.

Scotese, C.R., 1984, Paleozoic paleomagnetism in the assembly of Pangea, *in* Van der Voo, R., Scotese, C..R., and Bonhommet, N., eds., Plate reconstruction from Paleozoic magnetism: American Geophysical Union Geodynamics Series, v. 2, p. 1–10.

Secor, D.T., Jr., Samson, S.L., Snoke, A.W., and Palmer, A.R., 1983, Confirmation of the Carolina Slate Belt as an exotic terrane: Science, v. 221, p. 649–651.

Seiders, V.M., 1981, Geologic map of the Asheboro, North Carolina, and adjacent areas: U.S. Geological Survey Miscellaneous Investigations Map I-1314, scale 1:62 500.

Steiger, R.H., and Jäger, E., 1977, Subcommission on geochronology: Convention on the use of decay constants in geo- and cosmochronology: Earth and Planetary Science Letters, v. 36, p. 359–362.

Stromquist, A.A., and Sundelius, H.W., 1969, Stratigraphy of the Albemarle Group of the Carolina slate belt in central North Carolina: U.S. Geological Survey Bulletin 1274-B, p. 1–22.

Stromquist, A.A., Choquette, P.W., and Sundelius, H.W., 1971, Geologic map of the Denton quadrangle, central North Carolina: U.S. Geological Survey Quadrangle Map GQ-872, scale 1:62 500.

Sundelius, H.W., 1970, The Carolina Slate Belt, *in* Fisher, G.W., et al., eds., Studies of Appalachian geology: Central and southern: New York, Interscience, p. 351–367.

Sundelius, H.W., and Stromquist, A.A., 1978, Interpretive geologic map of the bedrock, Mount Pleasant quadrangle, Cabarrus and Stanly counties, North Carolina: U.S. Geological Survey Miscellaneous Investigations Map I-1082, scale 1:48 000.

Sutter, J.F., and Smith, T.E., 1979, $^{40}Ar/^{39}Ar$ ages of diabase intrusions from Newark trend basins in Connecticut and Maryland: American Journal of Science, v. 279, p. 808–831.

Van der Voo, R., and French, R.B., 1977, Paleomagnetism of the Late Ordovician Juniata Formation and the remagnetization hypothesis: Journal of Geophysical Research, v. 82, p. 5796–5802.

Van der Voo, R., and Johnson, R.J.E., 1985, Paleomagnetism of the Dunn Point Formation (Nova Scotia): High paleolatitudes for the Avalon terrane in the Late Ordovician: Geophysical Research Letters, v. 12, p. 337–340.

Van der Voo, R., French, R.B., and Williams, D.W., 1976, Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America: Journal of Geophysical Research, v. 81, p. 5633–5638.

Vick, H., Channell, J.E.T., and Opdyke, N.D., 1985, Paleomagnetism of low grade metamorphic rocks from the Carolina Slate Belt [abs.]: EOS (American Geophysical Union Transactions), v. 66, p. 258.

Watts, D.R., and Van der Voo, R., 1979, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Valley and Ridge province, eastern Tennessee: Journal of Geophysical Research, v. 84, p. 645–655.

Williams, H., and Hatcher, R.D., Jr., 1982, Suspect terranes and accretionary history of the Appalachian orogen: Geology, v. 10, p. 530–536.

Wright, J.E., and Seiders, V.M., 1980, Age of zircon from volcanic rocks of the central North Carolina Piedmont and tectonic implications for the Carolina volcanic slate belt: Geological Society of America Bulletin, v. 91, p. 287–294.

Zen, E-an, 1983, Exotic terranes in the New England Appalachians—Limits, candidates, and ages: A speculative essay, *in* Hatcher, R.D., Jr., Williams, H., and Zietz, I., eds., Contributions to the tectonics and geophysics of mountain chains: Geological Society of America Memoir 158, p. 55–81.

ACKNOWLEDGMENTS

Parts of this research were funded by a Geological Society of America grant to Noel and by a University of Georgia Faculty Research Grant to Spariosu.

Manuscript received November 17, 1986
Revised manuscript received September 29, 1987
Manuscript accepted October 20, 1987